M13-C3 learning-stage analysis six-cell AAMAS + ECAI on `m13_bucket3_binary_collider_and`

H0 closed / no Bucket 3 claim

Status. This document is the **closed H0** learning-stage record for the six target-wise cells on frozen sample `n30_seed42`. **No Bucket 3 claim** is approved. Markdown twin: `learning_analysis.md`. Completed follow-up probes H1–H3 are documented in `h1_support_ablation.md` and `h2_irrelevant_covariate.md`, and `h3_bk_leading_distractor.md` (not part of H0). H4/H5 status: `future_probes.md`. Pre-run dossier: `fixture_dossier.tex`.

1 Contract

Fixture `m13_bucket3_binary_collider_and` has DAG $A \to C \leftarrow B$, with $A \sim$ Bernoulli(4/5) and $B \sim$ Bernoulli(7/10) independent, and with deterministic child $C = A \wedge B$. The frozen sample has size n=30 and seed 42. Encoding is exact-value BK with binary 1-vs-0 examples. Both arms use brave learning and `check_ic`. The AAMAS arm is `aamas2025` (greedy / mgr / bk). The ECAI arm is `ecai2024` (nd / any / all). The evaluator-only reference for c is `c(A) :- a_val_1(A), b_val_1(A).`. Roots are marked `no_observed_parent_deterministic_rule`.

Collection roots lie under `causal/outputs/aba_learning/targetwise/m13_bucket3_binary_collider_and/{aamas2025,ecai2024}/n30_seed42/`. Arm `summary.md` files are the authoritative outcome tables.

2 Geometry for learning

Sample counts are $(1,1,1) : 17$, $(1,0,0) : 7$, $(0,1,0) : 4$, and $(0,0,0) : 2$ (rows 26 and 30 only). The implied $E^{\pm}$ sizes are a 24/6, b 21/9, and c 17/13.

Under $C = A \wedge B$, every positive example for c lies in the single predictor family $(A, B) = (1, 1)$. Positive examples for root a split into $(B, C) = (1, 1)$ and $(B, C) = (0, 0)$ (and symmetrically for b). The cell $(\textit{other root}, C) = (0, 0)$ is label-ambiguous. For target a, row 9 with values $(1, 0, 0)$ is E^+, while row 26 with values $(0, 0, 0)$ is E^-, yet both share the BK literals `b_val_0` and `c_val_0`. Roots have no deterministic causal rule over the other observed variables. A syntactic match between a delta and the evaluator reference for c is not automatic causal success.

3 Outcomes (verified)

Arm	Target	Outcome / surface delta
AAMAS	a, b	`completed_no_solution`
AAMAS	c	`solved`: `c :- a_val_1, b_val_1`
ECAI	a, b	`solved`: 11-clause α-framework
ECAI	c	`solved`: `c :-` α_1`, a_val_1` with contrary `b_val_0`

4 AAMAS narrative

The AAMAS arm uses `configs/aamas2025_config.pl`. The active options for these cells are `learning_mode(brave)`, `folding_mode(greedy)`, `folding_selection(mgr)`, `folding_space(bk)`, `asm_intro(relto)`, and `check_ic`. Under greedy folding with BK space, each positive example is replaced by a **maximal co-occurrence body** assembled from background predicates that hold of that row. The mgr selection step then keeps a most-general representative among duplicate bodies. Brave entailment is the acceptance test for each candidate extension. Authoritative traces are the per-cell `prolog.stdout` files under `.../aamas2025/n30_seed42/cells/target-{a,b,c}/output/`, together with the arm `summary.md`.

4.1 Target c — solved with an assumption-free conjunction

For target c, every positive example lies in the single joint cell $(A, B) = (1, 1)$. There are seventeen such rows, and each therefore admits the same maximal BK body $\{$`a_val_1`$, $`b_val_1`$\}$. After mgr deduplication, the learner retains one live candidate:

```
c(A) :- a_val_1(A), b_val_1(A).
```

Brave entailment of $\langle E^+, E^- \rangle$ succeeds immediately. No negative example shares that body, so no assumption is introduced. The folding queue is empty, and the run terminates as `solved` with zero assumptions and zero contraries. The post-run artefact audit on `bk.sol_chk.asp` reports SAT.

The emitted delta string is identical to the evaluator-only positive-state reference for c in `mechanism_reference.json`. That match is recorded here as a **syntactic coincidence**. It is not by itself a causal-recovery verdict. The same configuration and sample can fail on the root targets, as the next subsection shows.

Authoritative artefacts: `.../aamas2025/n30_seed42/cells/target-c/output/{delta.aba,prolog.stdout}` and the arm `summary.md`.

4.2 Targets a/b — `completed_no_solution` (symmetric)

The story for root a is verified in `.../aamas2025/n30_seed42/cells/target-a/output/prolog.stdout`. Target b is the same procedure with the roles of A and B swapped.

Positive examples for a fall into two BK co-occurrence families. The first is $(B, C) = (1, 1)$, which arises on every row where $A = 1$ and the AND child fires. The second is $(B, C) = (0, 0)$, which arises on rows such as row 9 where $A = 1$ yet $B = C = 0$. Greedy folding therefore proposes two distinct bodies.

The both-1 rule

```
a(A) :- b_val_1(A), c_val_1(A).
```

is accepted under brave entailment. It covers the $(1, 1)$ positives without touching any negative example.

The both-0 rule

```
a(A) :- b_val_0(A), c_val_0(A).
```

is then proposed from an early both-0 positive (the trace folds row 8). That body is **not** a function of the root. It also holds of the negative examples in rows **26** and **30**, which are the only sample atoms $(0, 0, 0)$. Brave entailment therefore fails. The learner introduces an assumption α_1 relative to the offending body, yielding

```
a(A) :- alpha_1(A), b_val_0(A), c_val_0(A).
```

Contrary rote learning then cites exactly rows 26 and 30 as grounds for `c_alpha_1`. Greedy folding of those contrary examples reconstructs the **same** both-0 body for the contrary,

```
c_alpha_1(A) :- b_val_0(A), c_val_0(A).
```

Under `asm_intro(relto)`, a further assumption cannot be introduced for that body because α_1 is already present. The engine reports KO and ends with `* No solution found!`. The cell outcome is `completed_no_solution`. No `delta.aba` and no `bk.sol.*` artefacts are emitted.

This is a **search and repair dead-end** on this example geometry under greedy, mgr, and relto. It is not a runner failure. The folding procedure is the same as for target c. What differs is the split of E^+ across two predictor cells and the presence of opposite labels inside the both-0 cell.

5 ECAI narrative

The ECAI arm uses `configs/ecai2024_config.pl`. The active options are `learning_mode(brave)`, `folding_mode(nd)`, `folding_selection(any)`, `folding_space(all)`, `asm_intro(relto)`, and `check_ic`. Unlike AAMAS, nd folding takes **one** predicate replacement per fold step and does not build a maximal co-occurrence catalogue. The first BK predicate that matches a selected example is typically the one that appears earliest in the declared predictor order. Authoritative traces are under `.../ecai2024/n30_seed42/cells/target-{a,b,c}/output/`.

5.1 Target c — solved by under-folding and a contrary

The path is verified in `.../ecai2024/n30_seed42/cells/target-c/output/prolog.stdout` and in the corresponding `delta.aba`.

Learning begins from rote positive rules indexed by sample id. The first selected positive is row 1. Nd folding replaces that id by `a_val_1`, which is the first BK predictor under the declared order a, then b. The resulting unary rule is

```
c(A) :- a_val_1(A).
```

That body is too strong for the AND child. Every row with $A = 1$ and $B = 0$ is an E^- example for c, yet satisfies `a_val_1`. Brave entailment fails, and the learner introduces α_1, giving

```
c(A) :- alpha_1(A), a_val_1(A).
```

Contrary rote learning then records `c_alpha_1` on the seven $(1, 0, 0)$ negatives that remain uncovered attacks, namely rows **8, 9, 13, 14, 18, 24,** and **25**.

The first contrary fold attempt replaces row 8 by `a_val_1`. That candidate fails, and relto cannot introduce a second assumption for that body. A subsequent fold replaces the same contrary example by `b_val_0`. Brave entailment then succeeds. All remaining contrary rotes are subsumed, and the queue empties.

The final delta is

```
c(A) :- alpha_1(A), a_val_1(A).
c_alpha_1(A) :- b_val_0(A).
assumption(alpha_1(A)).
contrary(alpha_1(A),c_alpha_1(A)) :- assumption(alpha_1(A)).
```

On this sample, the framework behaves extensionally like "derive c when $a = 1$, unless the assumption is attacked when $b = 0$". Under brave checking of the observed $E^{\pm}$, that presentation is aligned with $C = A \wedge B$. The ABA **shape** is nevertheless different from the AAMAS conjunction. In particular, the delta string does **not** match the evaluator reference `c(A) :- a_val_1(A), b_val_1(A).`. The run reports `solved` with artefact-audit SAT.

5.2 Targets a/b — solved by a brave choice gadget

The path for root a is verified in `.../ecai2024/n30_seed42/cells/target-a/output/{prolog.stdout,delta.aba}`. Target b is the mirror with $a \leftrightarrow b$. The arm summary records eleven delta rules, three assumptions, and three contraries for each root, with artefact-audit SAT.

Nd folding again takes one step at a time. From an early $(B, C) = (1, 1)$ positive (row 1), the first fold yields the unary body `b_val_1`. That rule over-generalises onto E^- rows where $B = 1$ and $C = 0$ (the trace cites rows 3, 5, 15, and 19 for the first contrary). Assumption α_1 is introduced, and the contrary later folds successfully to `c_val_0`. The resulting block separates the safe both-1 positives from the $B = 1, C = 0$ negatives:

```
a(A) :- alpha_1(A), b_val_1(A).
c_alpha_1(A) :- c_val_0(A).
```

A second ordinary target rule is then required for the remaining both-0 positives. Folding an early both-0 positive (row 8) yields `b_val_0`. Brave entailment fails because the same BK literals hold of the E^- atoms in rows **26** and **30**. Assumption α_2 is introduced on that body. Contrary rote cites those two negatives. Folding the contrary first attempts `b_val_0` (KO under relto against α_2), then `c_val_0`. Even the unary `c_val_0` contrary is still too strong for brave entailment of the joint example set, so a further assumption α_3 appears on the contrary. Rote learning of `c_alpha_3` then cites the both-0 **positives** (rows 8, 9, 13, 14, 18, 24, and 25). That contrary folds by re-using α_2 together with `b_val_0`. The resulting nest is

```
a(A) :- alpha_2(A), b_val_0(A).
c_alpha_2(A) :- alpha_3(A), c_val_0(A).
c_alpha_3(A) :- alpha_2(A), b_val_0(A).
```

together with the corresponding `assumption` and `contrary` declarations.

The important geometric point is the ambiguous cell $(B, C) = (0, 0)$. Row **9** is an E^+ example for a with joint values $(1, 0, 0)$. Row **26** is an E^- example with joint values $(0, 0, 0)$. Both rows share the BK literals `b_val_0` and `c_val_0`. No deterministic rule over (B, C) can assign both labels. The α_2/α_3 nest is a **per-row assumption choice gadget**. Within one witnessing stable model, the ground assumption atoms for different row identifiers can take different statuses, permitting a to hold at row 9 while being rejected at row 26. Other stable models may make different choices. Brave entailment requires the existence of at least one stable model that meets the **joint** $E^{\pm}$ constraints; it does not use a separate witness for each example and does **not** require a unique determination of a from (b, c).

The runner outcome `solved` therefore means that a serialized ABA framework passed brave entailment and that the artefact audit reported SAT. That outcome is **misleading** if it is read as recovery of a functional root mechanism. The evaluator reference for roots remains `no_observed_parent_deterministic_rule`.

6 Synthesis

	Child c	Roots a/b
AAMAS	Solved with an assumption-free AND conjunction. The delta string coincides with the evaluator reference.	No solution. The greedy both-0 body and contrary repair fail on rows 26 and 30.
ECAI	Solved with α_1 and a `b_val_0` contrary after under-folding to a.	Solved with a brave choice gadget on the ambiguous $(0, 0)$ cell.

For the roots, AAMAS fails without emitting a false functional both-0 Horn rule, and fails entirely. ECAI emits a non-functional brave covering of the same ambiguous cell. Neither recovers a root mechanism. All six cells use the same frozen table. The differences are strategy, entailment semantics, and target geometry, not resampling. The `sol_chk` SAT result is an artefact-integrity audit, not a separate coverage metric.

7 Non-claims

This write-up is not Russo-style Causal ABA, DAG recovery, or CPDAG recovery. It does not treat the AAMAS c string match as mechanism recovery. It does not treat the ECAI assumed form as "recovered AND" as a project claim. It does not parent-set score root deltas without the brave / non-functional caveat. Future probes H1–H5 are outside this H0 record. H1–H3 are documented in `h1_support_ablation.md`, `h2_irrelevant_covariate.md`, and `h3_bk_leading_distractor.md`. Remaining probe: `future_probes.md`.

8 Next

H0 is closed. Probes H1–H3 are documented in `h1_support_ablation.md`, `h2_irrelevant_covariate.md`, and `h3_bk_leading_distractor.md`. H4 infrastructure-ready/unrun and H5 deferred are recorded in `future_probes.md`. Still no Bucket 3 claim.